

Cybersecurity Threats and Defense

Cybersecurity protects systems, networks, and programs from digital attacks aimed at accessing, changing, or destroying data.

Common threats include malware, phishing, ransomware, denial-of-service attacks, and SQL injection vulnerabilities.

Zero Trust security models assume no user or system should be trusted by default, requiring verification at every access point.

Multi-factor authentication, encryption, and intrusion detection systems form the foundation of modern security defense.

Security Operations and Compliance

Security Operations Centers (SOCs) monitor and analyze organization security posture around the clock.

GDPR, HIPAA, and SOC 2 are major compliance frameworks that organizations must adhere to for data protection.

Penetration testing simulates cyberattacks to identify vulnerabilities before malicious actors exploit them.

Incident response plans outline procedures for detecting, containing, and recovering from security breaches.